

Fuzzy Systems

Question Bank

Optimization Techniques • CSE(DS)-5003 • Module 3

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Note to Students

This question bank is based entirely on the course notes *Fuzzy Systems: A Simple Introduction* distributed in class for Optimization Techniques (CSE(DS)-5003), Module 3. All definitions, examples, notation, and numerical values used in the questions are drawn directly from those notes.

The running example — academic performance assessment using fuzzy sets (Poor, Average, Good) over marks $x \in [0, 100]$ — appears throughout the questions. Attempt the questions on your own before checking the answer key.

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Multiple Choice Questions (MCQ)

Choose the single best answer for each question.

Introduction to Fuzzy Systems

Q1. Fuzzy logic was introduced by:

- (a) John Holland
- (b) John Koza
- (c) Lotfi Zadeh
- (d) Charles Darwin

Q2. In classical (crisp) logic, a membership value can only be:

- (a) Any real number
- (b) Any value in $[0, 1]$
- (c) Either 0 or 1
- (d) Either -1 or $+1$

Q3. In fuzzy logic, a membership value can be:

- (a) Only 0 or 1
- (b) Any value in $[0, 1]$
- (c) Any positive real number
- (d) Any integer

Q4. Fuzzy logic was introduced to handle:

- (a) Only binary data
- (b) Vague, imprecise, or uncertain information
- (c) Only numerical optimization problems
- (d) Exact mathematical proofs

Q5. In the running example, a student with marks 59 would be classified by classical logic as:

- (a) Partially good
- (b) Almost good
- (c) Not good (since $59 < 60$)
- (d) Fully good

Q6. Which of the following is *not* an application of fuzzy systems?

- (a) Washing machine control
- (b) Student performance assessment
- (c) Sorting a list of integers
- (d) Medical diagnosis

Q7. Fuzzy logic is said to be close to human reasoning because:

- (a) It uses binary decisions
- (b) It allows partial membership and gradual transitions rather than sharp yes/no boundaries
- (c) It requires exact numerical inputs
- (d) It always produces a unique correct answer

Q8. The year in which fuzzy logic was introduced is:

- (a) 1945
- (b) 1955
- (c) 1965
- (d) 1975

Q9. Classical logic forces every statement into:

- (a) A range of values
- (b) One of two categories: true (1) or false (0)
- (c) A probability value
- (d) A linguistic category

Q10. Which of the following best describes a fuzzy system?

- (a) A system that uses exact crisp rules
- (b) A system that maps crisp inputs to crisp outputs using fuzzy sets and if-then rules
- (c) A system that only works with binary data
- (d) A system that requires a mathematical model of the process

Fuzzy Set Theory

Q11. A fuzzy set assigns each element a membership value in:

- (a) $\{0, 1\}$

- (b) $[0, 1]$
- (c) $[0, \infty)$
- (d) $(-\infty, \infty)$

Q12. The membership function $\mu_A(x) = 1$ means:

- (a) x does not belong to A
- (b) x partially belongs to A
- (c) x fully belongs to A
- (d) x belongs to the complement of A

Q13. The membership function $\mu_A(x) = 0$ means:

- (a) x fully belongs to A
- (b) x partially belongs to A
- (c) x does not belong to A at all
- (d) x belongs to all fuzzy sets equally

Q14. A triangular membership function is defined by:

- (a) One point (the peak)
- (b) Two points (the feet)
- (c) Three points (a, b, c)
- (d) Four points (a, b, c, d)

Q15. In the triangular membership function with parameters (a, b, c) , the peak (membership = 1) occurs at:

- (a) $x = a$
- (b) $x = b$
- (c) $x = c$
- (d) $x = (a + c)/2$

Q16. Using the running example (Poor: $(0, 0, 50)$), the membership of a student with marks $x = 40$ in the Poor set is:

- (a) 0.00
- (b) 0.20
- (c) 0.50
- (d) 0.80

- Q17.** Using the running example (Average: (30, 50, 70)), the membership of a student with marks $x = 40$ in the Average set is:
- (a) 0.20
 - (b) 0.30
 - (c) 0.50
 - (d) 0.70
- Q18.** Using the running example (Good: (50, 100, 100)), the membership of a student with marks $x = 40$ in the Good set is:
- (a) 0.00
 - (b) 0.20
 - (c) 0.50
 - (d) 1.00
- Q19.** A trapezoidal membership function is defined by:
- (a) Two points
 - (b) Three points
 - (c) Four points (a, b, c, d)
 - (d) Five points
- Q20.** The trapezoidal membership function has membership = 1 for all x in the range:
- (a) $[a, b]$
 - (b) $[b, c]$
 - (c) $[a, d]$
 - (d) $[c, d]$
- Q21.** The Gaussian membership function is defined by:
- (a) Three points (a, b, c)
 - (b) Centre c and width σ
 - (c) Four points (a, b, c, d)
 - (d) Two feet and one peak
- Q22.** The fuzzy *union* of sets A and B is computed as:
- (a) $\mu_A(x) + \mu_B(x)$
 - (b) $\min(\mu_A(x), \mu_B(x))$
 - (c) $\max(\mu_A(x), \mu_B(x))$

(d) $\mu_A(x) \times \mu_B(x)$

Q23. The fuzzy *intersection* of sets A and B is computed as:

(a) $\max(\mu_A(x), \mu_B(x))$

(b) $\min(\mu_A(x), \mu_B(x))$

(c) $1 - \mu_A(x)$

(d) $\mu_A(x) \times \mu_B(x)$

Q24. The fuzzy *complement* of set A is:

(a) $\max(\mu_A(x), 0)$

(b) $\mu_A(x)/2$

(c) $1 - \mu_A(x)$

(d) $\min(\mu_A(x), 1)$

Q25. For $\mu_{\text{Poor}}(40) = 0.20$ and $\mu_{\text{Average}}(40) = 0.50$, the union (Poor OR Average) gives:

(a) 0.10

(b) 0.20

(c) 0.50

(d) 0.70

Q26. For $\mu_{\text{Poor}}(40) = 0.20$ and $\mu_{\text{Average}}(40) = 0.50$, the intersection (Poor AND Average) gives:

(a) 0.10

(b) 0.20

(c) 0.50

(d) 0.70

Q27. For $\mu_{\text{Poor}}(40) = 0.20$, the complement (NOT Poor) gives:

(a) 0.20

(b) 0.50

(c) 0.80

(d) 1.00

Q28. A *linguistic variable* is a variable whose values are:

(a) Only integers

(b) Words or linguistic terms (e.g. Poor, Average, Good)

- (c) Only real numbers
- (d) Only binary strings

Q29. In the running example, the linguistic variable is:

- (a) Marks
- (b) Grade point
- (c) Academic Performance
- (d) Membership value

Q30. The universe of discourse for the running example is:

- (a) {Poor, Average, Good}
- (b) $[0, 1]$
- (c) $[0, 100]$ (marks out of 100)
- (d) $[0, 10]$

Fuzzy Inference System

Q31. A Fuzzy Inference System (FIS) maps:

- (a) Fuzzy inputs to fuzzy outputs only
- (b) Crisp inputs to crisp outputs using fuzzy logic
- (c) Crisp inputs to fuzzy outputs only
- (d) Binary inputs to binary outputs

Q32. The first step of a FIS is:

- (a) Defuzzification
- (b) Aggregation
- (c) Fuzzification
- (d) Rule evaluation

Q33. Fuzzification converts:

- (a) A fuzzy set into a crisp number
- (b) A crisp input into fuzzy membership values
- (c) Rules into membership functions
- (d) Output sets into input sets

Q34. In the running example, fuzzification of marks $x = 40$ gives $\mu_{\text{Average}}(40) =$:

- (a) 0.00
- (b) 0.20
- (c) 0.50
- (d) 1.00

Q35. A fuzzy if-then rule has:

- (a) Only a consequent (THEN part)
- (b) Only an antecedent (IF part)
- (c) An antecedent (IF) and a consequent (THEN)
- (d) A loop and a condition

Q36. In the Mamdani method, the firing strength of a rule equals:

- (a) The maximum membership across all rules
- (b) The membership value of the rule's antecedent
- (c) The centroid of the output set
- (d) The average fitness of the population

Q37. For marks $x = 40$, Rule R1 (IF Poor THEN Fail) fires with strength:

- (a) 0.00
- (b) 0.20
- (c) 0.50
- (d) 1.00

Q38. For marks $x = 40$, Rule R2 (IF Average THEN Pass) fires with strength:

- (a) 0.00
- (b) 0.20
- (c) 0.50
- (d) 1.00

Q39. For marks $x = 40$, Rule R3 (IF Good THEN Distinction) fires with strength:

- (a) 0.00
- (b) 0.20
- (c) 0.50
- (d) 1.00

Q40. Aggregation combines the clipped rule outputs using:

- (a) The min operation
- (b) The max (union) operation
- (c) The average
- (d) The product

Q41. Defuzzification converts:

- (a) A crisp number into a fuzzy set
- (b) Membership functions into rules
- (c) An aggregated fuzzy set into a crisp output
- (d) Linguistic variables into chromosomes

Q42. The centroid defuzzification formula is:

- (a) $y^* = \max_i \mu(y_i)$
- (b) $y^* = \frac{\sum_i y_i \cdot \mu(y_i)}{\sum_i \mu(y_i)}$
- (c) $y^* = \min_i y_i$
- (d) $y^* = \frac{\sum_i y_i}{n}$

Q43. Using the aggregated output $\{(20, 0.20), (50, 0.50), (90, 0.00)\}$, the centroid defuzzification gives:

- (a) 35.00
- (b) 38.57
- (c) 41.43
- (d) 50.00

Q44. The centroid output of 41.43 for marks $x = 40$ reflects that the student is:

- (a) Fully Poor
- (b) Predominantly Average with a slight Poor component
- (c) Fully Good
- (d) Equally Poor and Good

Q45. The Mean of Maxima (MOM) defuzzification method:

- (a) Takes the centroid of the output set
- (b) Takes the average of all output values at maximum membership
- (c) Takes the minimum output value

- (d) Takes the maximum membership value directly
- Q46.** Which defuzzification method is preferred for most applications because it considers the full shape of the output set?
- (a) Smallest of Maxima
 - (b) Largest of Maxima
 - (c) Mean of Maxima
 - (d) Centroid method
- Q47.** The correct order of steps in a FIS is:
- (a) Defuzzification $\rightarrow$ Aggregation $\rightarrow$ Rule evaluation $\rightarrow$ Fuzzification
 - (b) Fuzzification $\rightarrow$ Rule evaluation $\rightarrow$ Aggregation $\rightarrow$ Defuzzification
 - (c) Rule evaluation $\rightarrow$ Fuzzification $\rightarrow$ Aggregation $\rightarrow$ Defuzzification
 - (d) Aggregation $\rightarrow$ Fuzzification $\rightarrow$ Rule evaluation $\rightarrow$ Defuzzification
- Q48.** In the Mamdani FIS, the output fuzzy set of each rule is clipped at:
- (a) The maximum membership across all rules
 - (b) The firing strength of that rule
 - (c) The centroid value
 - (d) The average membership of the input
- Q49.** If a rule fires with strength 0, its contribution to the aggregated output is:
- (a) Equal to the peak of its output set
 - (b) The centroid of its output set
 - (c) Zero (it does not activate)
 - (d) The complement of the output set
- Q50.** The number of rules in a FIS with one input variable and three linguistic values is:
- (a) 1
 - (b) 3
 - (c) 9
 - (d) 27

Optimization of Fuzzy Systems

- Q51.** The most common target when optimizing a fuzzy system with a GA is:

- (a) The defuzzification method
- (b) The membership function parameters
- (c) The population size
- (d) The crossover rate

Q52. When using a GA to optimize a fuzzy system, each chromosome encodes:

- (a) One membership value
- (b) One fuzzy rule
- (c) A set of membership function parameters
- (d) The centroid output

Q53. For optimizing three triangular membership functions (Poor, Average, Good), the chromosome contains how many parameters?

- (a) 3
- (b) 6
- (c) 9
- (d) 12

Q54. The fitness function used when optimizing a fuzzy system to minimize MSE is:

- (a) $F = \text{MSE}$
- (b) $F = 1 - \text{MSE}$
- (c) $F = 1/(1 + \text{MSE})$
- (d) $F = \sqrt{\text{MSE}}$

Q55. A perfect fuzzy system ($\text{MSE} = 0$) has fitness:

- (a) $F = 0$
- (b) $F = 0.5$
- (c) $F = 1$
- (d) $F = \infty$

Q56. In the running example, the initial MSE of 11.01 gives fitness:

- (a) 0.008
- (b) 0.083
- (c) 0.500
- (d) 0.917

- Q57.** After GA optimization in the running example, the MSE improved from 11.01 to 0.74. The new fitness is approximately:
- (a) 0.083
 - (b) 0.350
 - (c) 0.575
 - (d) 0.926
- Q58.** Real-coded encoding is preferred over binary encoding when optimizing fuzzy membership function parameters because:
- (a) The parameters are continuous values in $[0, 100]$
 - (b) Binary encoding is faster
 - (c) Real-coded encoding uses fewer chromosomes
 - (d) The parameters are always integers
- Q59.** Which of the following is *not* something that can be optimized in a fuzzy system using a GA?
- (a) Membership function parameters
 - (b) The rule base
 - (c) The number of fuzzy sets
 - (d) The universe of discourse (marks scale)
- Q60.** The main limitation of optimizing both rules and parameters simultaneously is:
- (a) It always produces a worse result
 - (b) The search space becomes very large
 - (c) It requires binary encoding only
 - (d) The fitness function cannot be defined

Short Answer Questions

Answer each question in 4–6 sentences.

- S1.** Explain the difference between a classical set and a fuzzy set. Why is a fuzzy set more appropriate for the student performance assessment problem?
- S2.** Define the triangular membership function. Using the running example (Average set: $a = 30, b = 50, c = 70$), compute $\mu_{\text{Average}}(x)$ for $x = 40$ and $x = 65$.
- S3.** Describe the four steps of a Fuzzy Inference System. Use the running example (marks $x = 40$) to illustrate each step briefly.
- S4.** What is the centroid defuzzification method? Using the aggregated output $\{(20, 0.20), (50, 0.50), (90, 0.30)\}$, compute the crisp output y^* .
- S5.** Explain how a Genetic Algorithm is used to optimize the membership function parameters of a fuzzy system. What does the chromosome encode and what is the fitness function?

Broad Questions

Answer each question in detail, with definitions, derivations, and examples where appropriate.

B1. Explain fuzzy set theory in detail. Define membership functions and describe three common types (triangular, trapezoidal, Gaussian). For the running example (student performance over marks $[0, 100]$), define three fuzzy sets (Poor, Average, Good) using triangular functions and compute membership values for a student with marks $x = 40$. Also compute and interpret the union, intersection, and complement of the Poor and Average sets at $x = 40$.

B2. Describe the complete working of a Fuzzy Inference System (FIS). Trace all four steps (fuzzification, rule evaluation, aggregation, defuzzification) for a student with marks $x = 40$, using the rules:

R1. IF Poor THEN Fail (output point 20)

R2. IF Average THEN Pass (output point 50)

R3. IF Good THEN Distinction (output point 90)

Compute the final crisp grade point using the centroid method and interpret the result.

B3. Discuss the advantages and limitations of fuzzy systems. Under what conditions would you prefer a fuzzy system over a classical crisp system? Give examples from the running example to support your answer.

B4. Explain how a Genetic Algorithm can be used to optimize a fuzzy system. Describe the chromosome encoding, fitness function, and GA loop for the running example (optimizing Poor, Average, Good membership functions to match faculty assessments). Show how fitness improves as MSE decreases from 11.01 to 0.74.

B5. Compare and contrast classical sets, fuzzy sets, and fuzzy inference systems. How do these three concepts build on each other to form a complete fuzzy decision-making system? Use the student performance assessment problem to illustrate each level.

Answer Key

MCQ Answers

Q	Topic	Ans	Q	Topic	Ans	Q	Topic	Ans	Q	Topic	Ans
1	Introduction	(c)	16	Fuzzy sets	(c)	31	FIS	(b)	46	FIS	(d)
2	Introduction	(c)	17	Fuzzy sets	(b)	32	FIS	(c)	47	FIS	(b)
3	Introduction	(b)	18	Fuzzy sets	(c)	33	FIS	(b)	48	FIS	(b)
4	Introduction	(b)	19	Fuzzy sets	(b)	34	FIS	(c)	49	FIS	(c)
5	Introduction	(c)	20	Fuzzy sets	(c)	35	FIS	(c)	50	FIS	(b)
6	Introduction	(c)	21	FIS	(b)	36	FIS	(b)	51	Optimization	(b)
7	Introduction	(b)	22	Fuzzy sets	(b)	37	FIS	(b)	52	Optimization	(c)
8	Introduction	(c)	23	Fuzzy sets	(c)	38	FIS	(c)	53	Optimization	(c)
9	Introduction	(b)	24	Fuzzy sets	(b)	39	FIS	(a)	54	Optimization	(c)
10	Introduction	(b)	25	Fuzzy sets	(b)	40	FIS	(b)	55	Optimization	(c)
11	Fuzzy sets	(b)	26	Fuzzy sets	(c)	41	FIS	(c)	56	Optimization	(b)
12	Fuzzy sets	(c)	27	Fuzzy sets	(b)	42	FIS	(b)	57	Optimization	(c)
13	Fuzzy sets	(c)	28	Fuzzy sets	(a)	43	FIS	(c)	58	Optimization	(a)
14	Fuzzy sets	(c)	29	Fuzzy sets	(b)	44	FIS	(b)	59	Optimization	(d)
15	Fuzzy sets	(b)	30	Fuzzy sets	(c)	45	FIS	(b)	60	Optimization	(b)

Table 1: MCQ Answer Key

Short Answer Model Answers

S1. Answer: A classical set assigns membership 0 or 1 to every element — either the element belongs or it does not. A fuzzy set assigns any value in $[0, 1]$, allowing partial membership. In the student performance problem, a student with marks 59 is almost good but classical logic would label them “not good” simply because $59 < 60$. A fuzzy set avoids this sharp, unrealistic boundary by assigning a high membership (e.g. 0.97) in the Good set to a student with marks 59, reflecting that they are nearly good. This is much closer to how teachers actually assess students.

S2. Answer: A triangular membership function is defined by three points (a, b, c) : the left foot a (membership rises from 0), the peak b (membership = 1), and the right foot c (membership falls back to 0). For the Average set ($a = 30, b = 50, c = 70$):

For $x = 40$ (between a and b):

$$\mu_{\text{Average}}(40) = \frac{40 - 30}{50 - 30} = \frac{10}{20} = 0.50$$

For $x = 65$ (between b and c):

$$\mu_{\text{Average}}(65) = \frac{70 - 65}{70 - 50} = \frac{5}{20} = 0.25$$

A student with marks 40 is 50% Average; a student with marks 65 is 25% Average (starting to move into the Good range).

S3. Answer: A FIS has four steps. (1) *Fuzzification*: for marks $x = 40$, compute $\mu_{\text{Poor}} = 0.20$, $\mu_{\text{Average}} = 0.50$, $\mu_{\text{Good}} = 0.00$. (2) *Rule evaluation*: R1 (Poor→Fail) fires at 0.20; R2 (Average→Pass) fires at 0.50; R3 (Good→Distinction) fires at 0.00. (3) *Aggregation*: combine clipped outputs using max: Fail at 0.20, Pass at 0.50, Distinction at 0.00. (4) *Defuzzification*: apply centroid to get $y^* \approx 41.43$, the final crisp grade point.

S4. Answer: The centroid method computes the weighted average of all output points, weighted by their membership values:

$$y^* = \frac{\sum_i y_i \cdot \mu(y_i)}{\sum_i \mu(y_i)}$$

Using the aggregated output $\{(20, 0.20), (50, 0.50), (90, 0.00)\}$:

$$y^* = \frac{20 \times 0.20 + 50 \times 0.50 + 90 \times 0.00}{0.20 + 0.50 + 0.00} = \frac{4 + 25 + 0}{0.70} = \frac{29}{0.70} \approx 41.43$$

The crisp output is 41.43, reflecting a mix of Average (dominant) and Poor (minor) performance.

S5. Answer: Each chromosome encodes the membership function parameters (a, b, c) for all three fuzzy sets, giving nine values: $[a_P, b_P, c_P, a_A, b_A, c_A, a_G, b_G, c_G]$. Real-coded

encoding is used since the parameters are continuous. The fitness function converts the minimisation objective (MSE) to a maximisation form: $F = 1/(1 + \text{MSE})$. The GA initialises a random population of parameter sets, evaluates each by running the full FIS on the training data and computing MSE, then applies selection, crossover, and mutation to evolve toward lower MSE. After T generations, the best chromosome gives the optimised membership function parameters.

Broad Question Model Answers

B1. Answer: A fuzzy set assigns each element x a membership value $\mu_A(x) \in [0, 1]$ rather than just 0 or 1.

Triangular (a, b, c) : rises linearly from a to peak at b , falls to c . *Trapezoidal* (a, b, c, d) : rises from a , flat top from b to c , falls to d . *Gaussian* (c, σ) : smooth bell curve, $\mu(x) = e^{-(x-c)^2/2\sigma^2}$.

Running example ($x = 40$):

$$\mu_{\text{Poor}}(40) = \frac{50 - 40}{50 - 0} = 0.20, \quad \mu_{\text{Average}}(40) = \frac{40 - 30}{50 - 30} = 0.50, \quad \mu_{\text{Good}}(40) = 0.00$$

Fuzzy operations: Union: $\max(0.20, 0.50) = 0.50$; Intersection: $\min(0.20, 0.50) = 0.20$; Complement (NOT Poor): $1 - 0.20 = 0.80$.

The student is 80% “not poor” — consistent with marks of 40.

B2. Answer: *Step 1 — Fuzzification* ($x = 40$): $\mu_P = 0.20$, $\mu_A = 0.50$, $\mu_G = 0.00$.

Step 2 — Rule evaluation: R1 fires at 0.20 $\Rightarrow$ Fail clipped at 0.20. R2 fires at 0.50 $\Rightarrow$ Pass clipped at 0.50. R3 fires at 0.00 $\Rightarrow$ Distinction not active.

Step 3 — Aggregation: Combine using max: $(20, 0.20)$, $(50, 0.50)$, $(90, 0.00)$.

Step 4 — Defuzzification (centroid):

$$y^* = \frac{20(0.20) + 50(0.50) + 90(0.00)}{0.20 + 0.50 + 0.00} = \frac{29}{0.70} \approx 41.43$$

The student receives a grade point of 41.43, reflecting predominantly Average performance with a small Poor component — a realistic and interpretable result for marks of 40.

B3. Answer: *Advantages:* handles imprecision naturally; close to human reasoning; produces smooth gradual outputs; no mathematical model required; robust to noise.

Limitations: membership function design is subjective; no default learning from data; rule explosion with many input variables; no guarantee of optimality.

When to prefer fuzzy: when the problem involves vague linguistic categories (Poor, Average, Good); when expert knowledge is available but data is limited; when interpretable rules are needed. In the running example, a fuzzy system is clearly better because students with marks 59 and 61 should receive similar assessments — classical logic treats them very differently.

When not to prefer fuzzy: when the problem is precisely defined and exact boundaries are appropriate; when a large training dataset is available and a machine learning model would perform better.

B4. Answer: The chromosome encodes nine parameters: $[a_P, b_P, c_P, a_A, b_A, c_A, a_G, b_G, c_G]$. Real-coded encoding is used. Fitness: $F = 1/(1 + \text{MSE})$.

Initial: parameters $(0, 0, 50)$, $(30, 50, 70)$, $(50, 100, 100)$ give $\text{MSE} = 11.01$, $F = 0.083$.

GA loop: select fitter chromosomes (roulette-wheel); crossover pairs of parameter sets; mutate with small perturbations; evaluate FIS on all four training students; repeat for T generations.

After 50 generations: parameters $(0, 0, 46)$, $(28, 52, 69)$, $(54, 100, 100)$ give $\text{MSE} = 0.74$, $F = 0.575$. Fitness improved from 0.083 to 0.575 — a 593% improvement. The boundaries shifted slightly to better match the faculty assessments.

B5. Answer: *Classical sets* (binary): membership is 0 or 1. A student either passes or fails. Sharp, unrealistic for gradual phenomena like academic performance.

Fuzzy sets (graded): membership is in $[0, 1]$. A student with marks 40 is 20% Poor and 50% Average simultaneously. More realistic and informative.

Fuzzy Inference System (decision making): combines fuzzy sets with if-then rules to produce a final crisp output. For marks 40, the FIS outputs grade point 41.43 — reflecting the dominant Average and minor Poor components, and matching the expected faculty assessment of 42 closely.

The three levels build naturally: fuzzy sets provide the language (Poor, Average, Good); the FIS provides the reasoning (rules and inference); optimization (GA) tunes the design to match real data. Together they form a complete, interpretable, and automatically tunable decision-making system.